

Cache Memory

What is Cache memory?

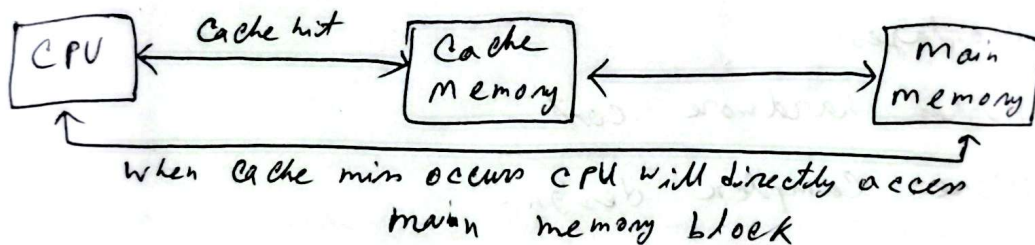
=> Cache memory stores frequently used instructions and data so ~~that~~ the CPU can access them faster than RAM.

=> Cache memory is a small, very fast memory located between CPU and main memory (RAM).

Purpose:

- Reduce CPU waiting time
- Improve system speed.

Working of Cache memory:



Two types of Cache Accesses:

- Simultaneous Cache Access.
- Hierarchical Cache Access.

Simultaneous Cache Access

In simultaneous access, cache and main memory are accessed at the same time.

- CPU sends address to cache and RAM simultaneously.
- if cache hit $\rightarrow$ Cache data is used, RAM result ignored.
- if cache miss $\rightarrow$ RAM data is used.

Here,

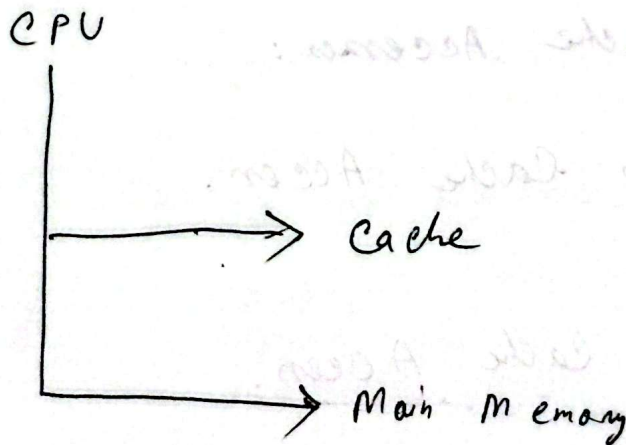
no waiting to check cache first and then decision.

Advantages:

- $\rightarrow$ Faster access time
- $\rightarrow$ Reduces delay on cache miss

Disadvantages:

- $\rightarrow$ Higher hardware cost
- $\rightarrow$ More complex design.



Hierarchical Cache Access:

In hierarchical access, memory is accessed level by level.

CPU $\rightarrow$ L1 $\rightarrow$ L2 $\rightarrow$ L3 $\rightarrow$ RAM

- CPU checks L1 cache
- If miss $\rightarrow$ check L2
- If miss $\rightarrow$ check L3
- Finally $\rightarrow$ main memory

Advantages:

- $\rightarrow$ Cost-effective
- $\rightarrow$ Efficient use of fast memory

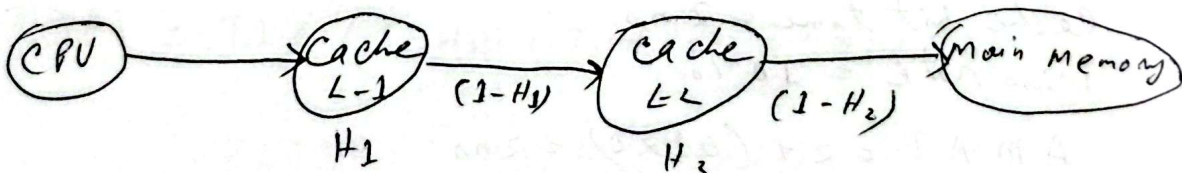
Disadvantages:

- $\rightarrow$ ~~slow~~ Slower on multiple cache misses.

Level 1:



Level 2:



Average Memory Access Time (AMAT):

AMAT is the average time needed to access data from memory considering hits and misses.

Basic AMAT Formula (single cache):

$$AMAT = \text{Hit time} + (\text{Miss Rate} \times \text{Miss penalty})$$

↳ Time to access cache
 Probability of cache miss
 Extra time to fetch from RAM

For Simultaneous Access:

Cache hit → Use cache result

Cache miss → RAM result already ready

Formula

$$AMAT = \text{Cache Hit}_{\text{time}} + (\text{Miss Rate} \times 0)$$

$$\approx \text{Cache Hit Time}$$

Cache hit time = 2 ns

Miss rate = 10%

$$AMAT = 2 + (0.1 \times 0) = 2 \text{ ns}$$

for more

Greeks for Greeks

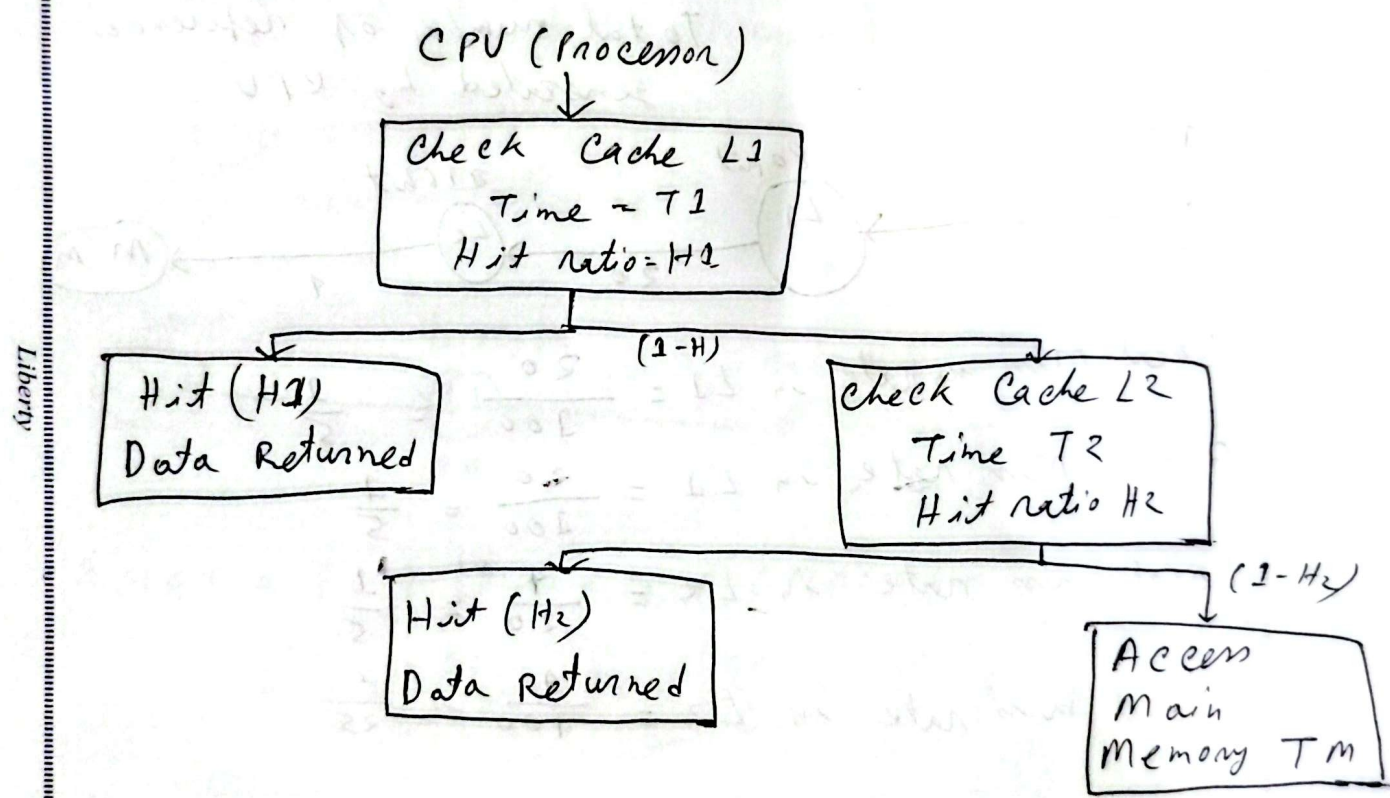
for Hierarchical Access:

Formula:

$$AMAT = \text{Hit ratio} * \text{Cache Memory Access Time} + (1 - \text{Hit ratio}) * (\text{Cache memory Access Time} + \text{Main Memory Access Time})$$

for Level 2

$$AMAT = T_1 + (1 - H_1) T_2 + (1 - H_1) (1 - H_2) T_m$$



$$AMAT = T_1 + (1 - H_1) (T_2 + (1 - H_2) T_m)$$

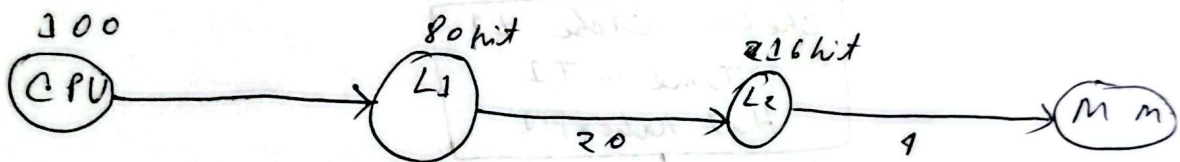
$$= H_1 T_1 + (1 - H_1) H_2 (T_1 + T_2) + (1 - H_1) (1 - H_2) (T_1 + T_2 + T_m)$$

Multilevel cache:

In multilevel cache, two kind of miss will calculate: -

$$\textcircled{1} \text{ Local Miss Rate} = \frac{\text{Number of miss in cache}}{\text{Number of references in that cache}}$$

$$\textcircled{2} \text{ Global Miss Rate} = \frac{\text{Number of miss in cache}}{\text{Total number of references generated by CPU}}$$



$$\text{Local miss rate in } L1 = \frac{20}{100} = \frac{1}{5}$$

$$\text{Global miss rate in } L1 = \frac{20}{100} = \frac{1}{5}$$

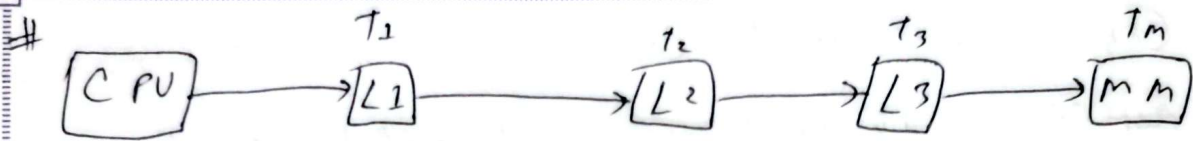
$$\text{Local miss rate in } L2 = \frac{4}{20} = \frac{1}{5}$$

$$\text{Global miss rate in } L2 = \frac{4}{100} = \frac{1}{25}$$

$$\boxed{\# \text{ GMRL2} = \text{LMRL1} \times \text{LMRL2}}$$

$$= \frac{1}{5} \times \frac{1}{5}$$

$$= \frac{1}{25}$$

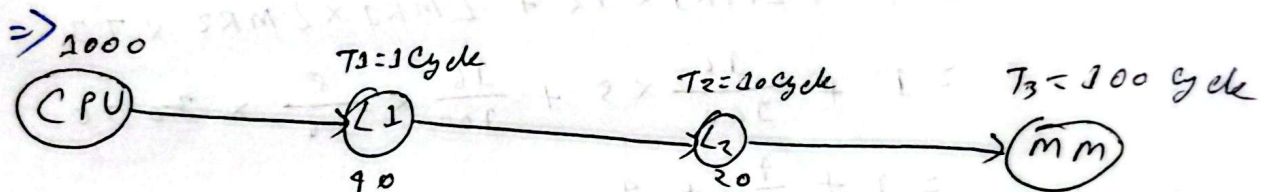


$$T_{avg} = T_1 + \frac{L_{MR1} \times T_2}{L_{MR1}} + \frac{L_{MR1} \times L_{MR2} \times T_3}{L_{MR1} \times L_{MR2}} + \frac{L_{MR1} \times L_{MR2} \times L_{MR3} \times T_{Am}}{L_{MR1} \times L_{MR2} \times L_{MR3}}$$

CPU generate 1000 reference. NO of min in L1 is 900, NO. of min in L2 is 20. If hit time of L1 is 1 cycle and hit time of L2 is 10 cycle.

i) Find AMAT,

ii) If miss penalty of L2 is 100 cycle, ~~AMAT~~ →



$$AMAT = T_1 + L_{MR1} \times T_2 + L_{MR1} \times L_{MR2} \times T_3$$

$$= 1 + \frac{90}{1000} \times 10 + \frac{90}{1000} \times \frac{20}{90} \times 100$$

$$= 1 + \frac{9}{10} + 2$$

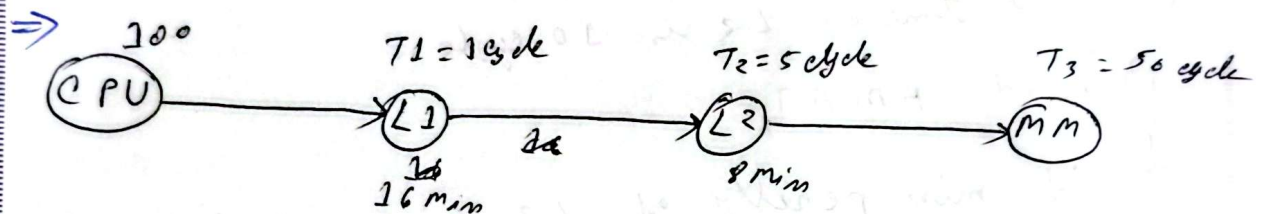
$$= 3.9 \text{ cycle}$$

Consider a 2 level cache system for 100 memory reference 16 miss in first level & 8 miss in second level.

Miss penalty from L2 to MM is 50 cycle

Hit time of L2 is 5 cycle & L1 is 1 cycle.

Find AMAT?



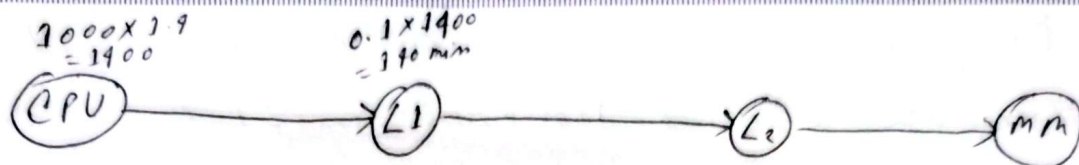
$$AMAT = T_1 + LMR_1 \times T_2 + LMR_1 \times LMR_2 \times T_3$$

$$= 1 + \frac{16}{100} \times 5 + \frac{16}{100} \times \frac{8}{16} \times 50$$

$$= 1 + \frac{4}{5} + 4$$

$$= 5.8 \text{ cycle}$$

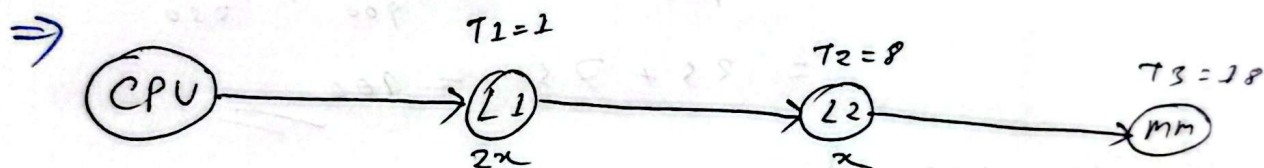
Consider a two-level cache hierarchy with L1 & L2 Cache. An application use 1.4 memory access/instruction on average. For this application, the miss rate of L1 cache 0.1, L2 cache experience on average 7 miss per ¹⁰⁰⁰ instruction. The miss rate L2 expressed correct to decimal place is ____.



Global miss rate of $L2 = 7$

$$\therefore \text{Local miss rate of } L2 = \frac{7}{190} = 0.05$$

In a 2-level Cache system the access time of $L1$ & $L2$ Cache are 1 & 8 clock cycle respectively. The miss penalty from $L2$ to main memory is 18 clock cycle. The miss rate of $L1$ is twice of $L2$. The average memory access time of this cache is 2 clock cycle. Miss rate of $L1$ & $L2$ is?



$$AMAT = T_1 + 2x \times 8 + 2x \times x \times 18$$

$$\Rightarrow 2 = 1 + 16x + 36x^2$$

$$\Rightarrow 36x^2 + 16x - 1 = 0$$

$$\Rightarrow 36x^2 + 18x - 2x - 1 = 0$$

$$\Rightarrow 18x(x + 1) - 1(2x + 1) = 0$$

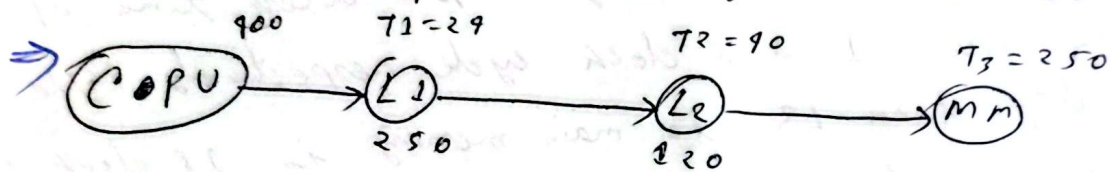
$$\Rightarrow (18x - 1)(x + 1) = 0$$

$$\therefore x = \frac{1}{18}, x = -\frac{1}{2}$$

$$L2 = x = \frac{1}{18}$$

$$L1 = 2x = \frac{1}{9}$$

During a program execution out of 1000 instructions, there are 250 & 120 misses in L1 & L2 cache respectively. Hit time for L1 & L2 are 29 & 90 cycle respectively. If there are 2.5 memory reference per instruction, how many average stall cycle per instruction if main memory access time 250 cycles.



?? NO. of memory reference = $\frac{1000}{2.5} = 400$

$$\text{Avg. stall cycle} = \frac{250}{400} \times 90 + \frac{250}{400} \times \frac{120}{250} \times 250$$

$$= 25 + 75 = 100$$

The access time of cache memory is 100ns and that of the main memory is 1 μ s. 80% of the memory requests are for reading and others are for write. The hit ratio for reading only access is 0.9. A write of the procedure is used. The average access time of the system for both the read and write request, is ... ?

⇒ Cache access time, $T_c = 100 \text{ ns}$

Main memory access time, $T_m = 1 \mu\text{s} = 1000 \text{ ns}$

Read request = 80%

∴ write request = 20%

Read hit ratio, $H_1 = 0.9$

From the write through of the procedure

$$T_{\text{write}} = T_c + T_m = 100 + 1000 = 1100 \text{ ns} = \text{AMAT}_{\text{write}}$$

$$\text{Read min time} = 100 + 1000 = 1100 \text{ ns}$$

For read,

$$\text{AMAT}_{\text{read}} = H_1 T_c + (1 - H_1) T$$

$$= 0.9 \times 100 + (1 - 0.9) \times 1100$$

$$= 200 \text{ ns}$$

$$\text{Overall AMAT} = (0.8 \times 200) + (0.2 \times 1100)$$

$$= 160 + 220 = 380 \text{ ns}$$

If the main memory access time is 100ns and cache access time is 50ns, Cache hit rate 90%, then what is AMAT to read from memory?

$$\Rightarrow \text{AMAT} = H_1 T_1 + (1 - H_1) T_2$$

$$= 0.9 \times 50 + (1 - 0.9) \times 100$$

$$= 45 + 10$$

$$= 55 \text{ ns}$$

In 2-level hierarchy, if the top level has an access time of 10 ns and the bottom level has an access time of 60 ns, what is the hit rate on the top level required to give an average access time of 15 ns?

$$\Rightarrow T_1 = 10 \text{ ns} \quad T_2 = 60 \text{ ns} \quad \text{AMAT} = 15 \text{ ns} \quad H_1 = ?$$

$$\therefore \text{AMAT} = H_1 T_1 + (1 - H_1) (T_1 + T_2)$$

$$\Rightarrow 15 = H_1 \times 10 + (1 - H_1) (10 + 60)$$

$$= 10 H_1 + 70 - 70 H_1$$

$$\Rightarrow 20 H_1 = 55$$

$$\therefore H_1 = \frac{55}{60} = 0.9167$$

Q? # In 2-level hierarchy, the top level has an access time of 10 ns and bottom level has an access time of 50 ns, the hit rate on the top level is 90%. If the block size of cache is 16 bytes, then what is the ~~average~~ AMAT required? (Consider the system uses locality of reference).

$$\Rightarrow \text{AMAT} = H_1 \times T_1 + (1 - H_1) ((\text{Block size} \times T_2) + T_1)$$

$$= 0.9 \times 10 + (1 - 0.9) ((16 \times 50) + 10)$$

$$= 9 + 0.1 (800 + 10) = 9 + 81 = 90 \text{ ns}$$

In 2-level hierarchy, the cache has an access time of 12 ns and the main memory access time of 120 ns, the hit rate of the cache is 90%. If the block size of cache is 16 bytes then what is the AMAT including miss penalty? (Miss penalty: Time to bring main memory block to cache memory when cache miss occurs)

$$\begin{aligned}
 \Rightarrow \text{AMAT} &= H_1 T_1 + (1 - H_1)(T_1 + (\text{Block size} * T_2) + \text{Cache memory access time due to block transfer}) \\
 &= 0.9 * 12 + (1 - 0.9)(12 + 16 * 120 + 12) \\
 &= 10.8 + 0.1(24 + 1920) = 10.8 + 194.4 \\
 &= 205.2 \text{ ns} //
 \end{aligned}$$